



Group 2:

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FLIGHT GAME

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Contents

1. INTRODUCTION	1
1.1 The Purpose	1
1.2 Target audience	1
1.3 The structure of the document	1
2. CURRENT STATUS	1
3. VISION	2
3.1. The general idea of game	2
3.2. How to play Game	2
4. FUNCTIONAL REQUIREMENTS	4
4.1. Functions	4
4.1.1. Start Game	4
4.1.1.1. Continuing a Game	4
4.1.1.2. Starting a New Game by creating an account	5
4.1.2. Start of the Journey	5
4.1.3. Complete Tasks to Progress	6
4.1.3.1. Start Mission - Answering Questions to Complete the Mission	7
4.1.3.2. Next mission	7
4.1.4. Creating the Team	7
4.1.5. Game result	7
5. QUALITY REQUIREMENTS	8

1. INTRODUCTION

1.1 The purpose

This document aims to edit and expand the requirements for developing a flight simulation game. This document will guide project team members, the teachers, and other class members in understanding the game's functions, quality requirements, and overall vision.

1.2 Target audience

This document is for:

1. Project development team members.
2. The teachers are interested in the progress of the project.
3. Classmates who wish to try out the game.

1.3 The structure of the document

The structure of the document consists of the following five parts:

1. Introduction
2. Current State
3. Vision
4. Functional Requirements
5. Quality Requirements

2. THE CURRENT STATUS

FLIGHTGAME v1.0 is an action-adventure game based on the story of a pilot who loves adventure and accidentally embarks on an adventure and searches for a gemstone through different lands by plane.

FLIGHTGAME v1.0 is currently a demo version with basic Python coding. It allows the users to:

- Input user name.
- Choose the type of plane.

- Select an airport.
- Complete tasks by answering questions.
- Calculating score.

FLIGHTGAME v1.0 has some limitations, including:

- The limited number of airports.
- Lack of diverse tasks.
- No graphics and user interface

3. VISION

3.1. The general idea of the game

FLIGHTGAME v2.0 continues to be based on the basic story where players take on the role of a pilot caught up in the quest for a precious gem that is the key to opening a treasure cave. Players will choose different airports based on the game's suggestions in this journey. At each airport that players choose, there will be a mandatory mission to gather information about the gem and increase the player's assets. Finding the gem depends on the information and assets the player collects at the airports.

The expanded FLIGHTGAME v2.0 will allow players to experience more realistic role-playing and have a user interface for players to interact within the browser. The game will use actual data about the location of airports worldwide, a world map, and an online weather service.

In addition, the missions will be classified by difficulty level according to the real weather at the airport.

3.2. How to play the game

1. The players create a new account with information: username, password, continent, country, start airport, plane.
2. Login as a player with username and password.
3. Complete mission to progress.
4. View current mission, progress.
5. View leaderboard and compete with other players.

The game's operation from start to end will be presented through a game flow diagram as follows:

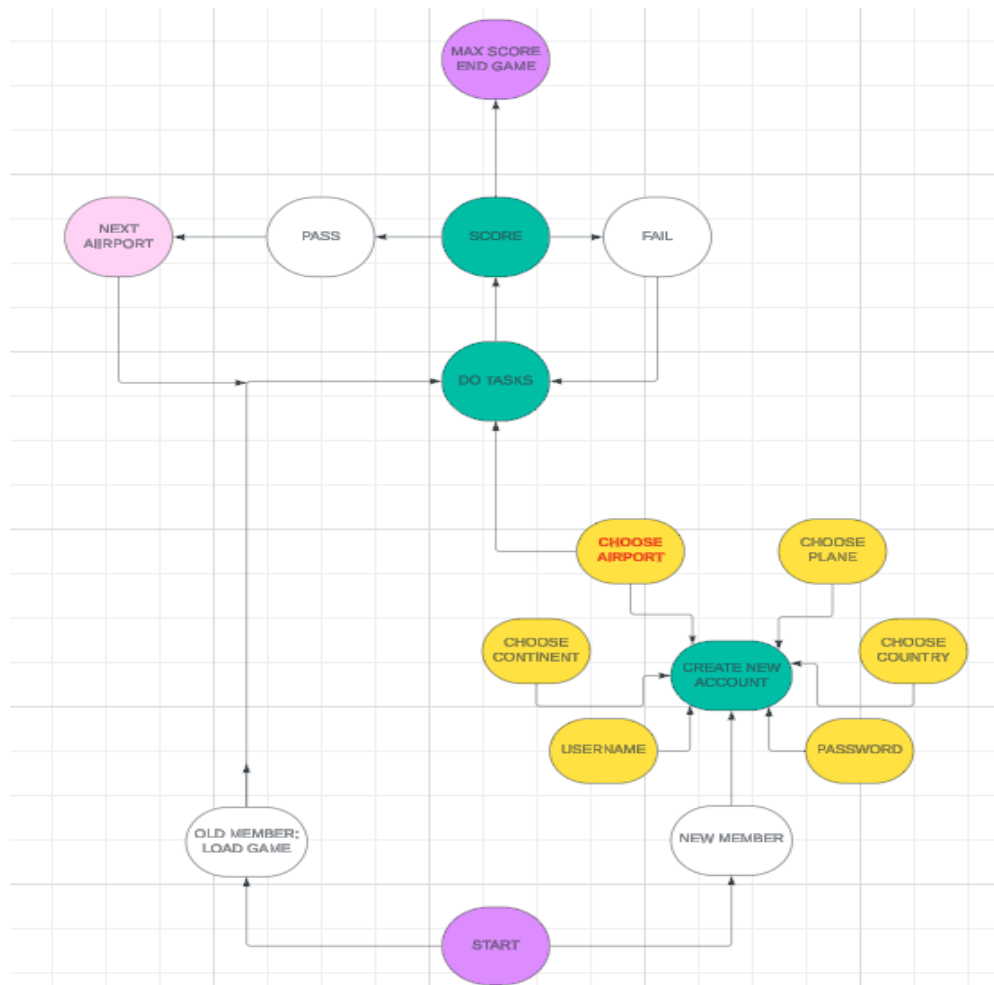


Figure 1: Game flowchart

4. FUNCTIONAL REQUIREMENTS

4.1. The functions

4.1.1. Starting the game

As a first-time or returning player, I want to see a welcoming introduction screen with options to Start a new game or Continue my previous journey, So that I can choose how to proceed based on my progress.

4.1.1.1. Continuing a Game

As a returning player, I want to log into my existing account by entering my username and password, so that I can continue my previous adventure from where I left off.

When the user clicks **"Continue My Journey"**, a login popup/modal appears.

- **Username (required):** Must match an existing account.
- **Password (required):** Must match the correct password for the username.

The system validates the form when the user clicks **"Continue My Journey"**:

- Ensure all fields are filled.
- Verify that the username and password combination exists in the database.
- Display an error message if validation fails

If validation succeeds, the system redirects the user to the game page. The system stores the player ID in localStorage for later use.

4.1.1.2. Starting a New Game by creating an account.

As a new player, I want to register for a new game by entering personal details (username, password) and selecting my starting location (continent, country, airport) and plane, so that I can create an account and begin my adventure in the game.

When the user clicks the **"Start New Journey"** button, a registration popup/modal appears.

- **Username (required):** Must be unique.
- **Password (required):** Must be secure.
- **Continent (required):** Dropdown list of continents.
- **Country (required):** Filtered list based on the selected continent.
- **Starting Airport (required):** Filtered list based on the selected country.
- **Plane Selection (required):** Display a selection of plane icons to choose from.

The system validates the form when the user clicks **"Start My Journey"**:

- Ensure all fields are filled.
- Check if the **username already exists** in the database.
- Display appropriate error messages if validation fails.

If validation succeeds, the system creates the player in the database and automatically redirects the user to the game page.

4.1.2. Start of the Journey

As a player logged into the game, I want to see my current mission displayed on a map, along with my game status and progress, so that I can track my adventure, check my scores, and continue my journey easily.

The page shows a map with all markers indicating all mission's location. The current mission marker is centered on the map.

Game Status Section displays essential game details:

- **Total Adventures:** Number of missions completed.
- **Total Score:** Total points earned.
- **Current Mission:** Name of the current mission's airport.
- A **"Continue Journey"** button

4.1.3. Complete Tasks to Progress.

As a player continuing the adventure, I want to see my current mission's location and weather details, along with a button to start the mission, so that I can understand where I am and prepare for the upcoming challenge.

When the user clicks "Continue Journey", the system replaces the map view with the mission information box to show the current mission's location name, current weather data fetched from the weather API (Weather Icon, Weather Condition and Temperature - Celsius) and "Start Mission" button.

4.1.3.1. Start Mission - Answering Questions to Complete the Mission

As a player continuing the adventure, I need to answer a set of Trivia questions based on the current location's weather, so that I can complete the mission and earn points to progress in the game.

When the user clicks "Start Mission", the system will fetches 5 random trivia questions from the OpenTrivia API based on the current weather conditions:

- Clear weather: Easy
- Cloudy or misty weather: Medium
- Rainy or snowy weather: Hard

Each question has multiple-choice options, the user can only select one answer per question. The player earns 10 points per correct answer (up to 50 points). After answering all 5 questions, the system checks the score:

- 3/5 or higher correct: the mission is completed successfully. The progress will be updated in the Game Status box and a new random airport with a new mission will be assigned to the user.

- Fewer than 3 correct: the mission fails. A message appears prompting the user to restart the mission. The system fetches another set of questions and restarts the mission.

4.1.3.2. Next mission

As a player who has successfully completed a mission, I want to receive a new random airport as the next mission location, so that I can continue exploring new destinations and progressing in the game.

After completing the current mission, the system will randomly select a new airport from available destinations and ensure that the selected airport has not been visited before. The system also updates the player's current location in the database, and a new record is added to the player progress table with a status of "Not Started".

4.1.4. Creating the team

As a player interested in team competition, I want to join or create a team if players are not in one, or view the team's details if players are already a member, so that I can compete with friends and track team progress.

When the player clicks the "My Team" tab, the system will check if the player belongs to a team using the database.

- If the player is not in a team, the system displays a "Join an Existing Team" button that lists available teams and a "Create a New Team" button with a form for entering a team name.
- If the player is in a team, the system displays the team name and list of team members, including: Player usernames, Total adventures completed and Total score earned.

4.1.5. Game result

As a player, I want to see the top 5 players ranked by their total score, so that I can track the highest-ranking players and compare my progress.

When the player navigates to the Leaderboard page, the system retrieves the top 5 players with the highest total scores from the database. The page will display Player's rank (1-5), Player's username, Total score and Total adventures.

5. QUALITY REQUIREMENTS

- Fetching airport information from the database must take a maximum of two seconds.
- Fetching data and displaying the map using LeafletJs.
- Fetching weather data using OpenWeatherMap API.
- Fetching random questions using OpenTrivia API based on weather conditions.
- The user must get instant feedback from all actions they perform.
- The score (point) calculation should account for every action (mission) that the player completes/fails.
- Clear feedback should be provided throughout the player's selection process from the game's beginning to the end.